Low-pass vs High-pass Filtering of Sentinel-2 Imagery

Which technique suits which remote sensing task?
EES403 Course Project

Group 1

Pranay Chakraborty (MS22041), Sangeeta Kumari (PH25084), Aparna Manoj (MS24070), Robin (MS23138), Sangam (MS23222)

Instructors: Dr. Chandrakant Ojha and Dr. Meenakshi Devi
Department of Earth and Environmental Sciences
Indian Institute of Science Education and Research, Mohali



We compared low-pass and high-pass filters on one Sentinel-2 scene of the Aravalli ridge between Gurugram and Faridabad. We implemented every filter ourselves, in the spatial and the frequency domain, and tested them on five measurable questions. The short answer: it depends on the task, and we will show the numbers that decide it.

Spatial frequency: how fast brightness changes across the ground

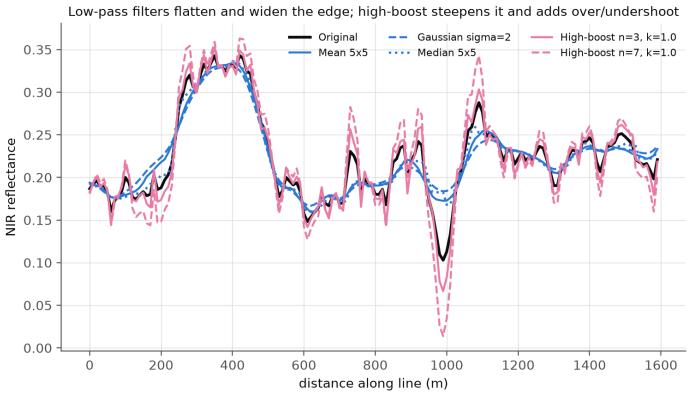
Low frequency: slow change
fields, water, broad slopes

High frequency: abrupt change
roads, roofs, ridge crests, noise

Low-pass (LPF): keep slow change
= smoothing

High-pass (HPF): keep abrupt change
HPF = original − LPF

High-boost: original + k·HPF
= sharpening

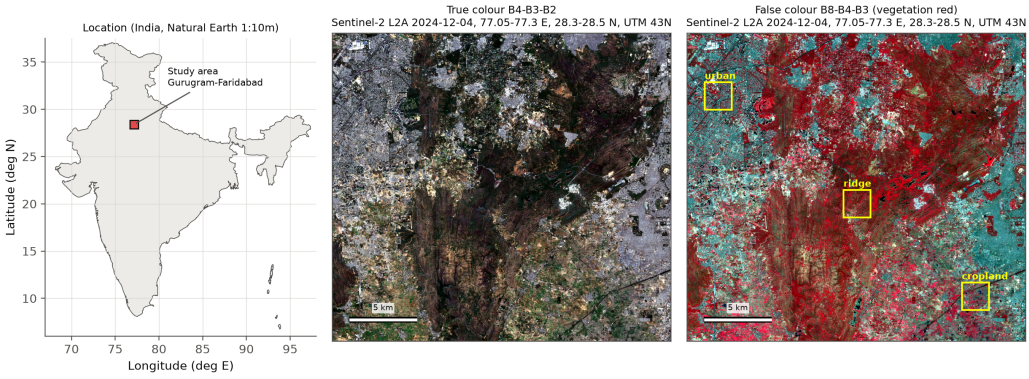


NIR reflectance along a 1.6 km line on the ridge.

Our question: which of the two is more useful, and *for which job*?

Frequency here means how often brightness changes as you move across the image. A field changes slowly, a street grid changes every few pixels. A low-pass filter averages neighbours, so slow changes survive and fast ones are smoothed away. A high-pass filter keeps only the fast changes; the textbook remote sensing version is the original minus a smoothed copy. High-boost adds that edge layer back. In the plot, the blue low-pass curves are flatter than the black original and the pink high-boost curves are steeper and overshoot.

Study area: ridges, city and farmland in one scene

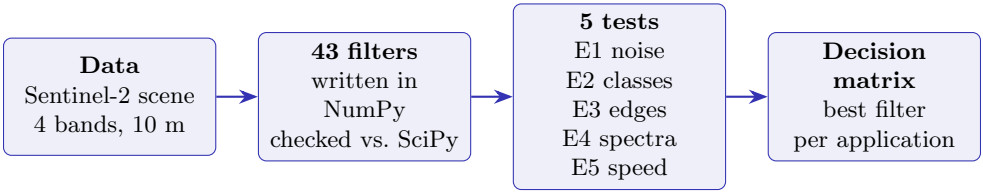


Area: Aravalli ridge, Gurugram–Faridabad, 25 × 23 km
Bands: blue, green, red, NIR at 10 m

Scene: Sentinel-2 L2A, 2024-12-04, best of 11
Reference: ESA WorldCover 2021 land cover

We chose this area because one scene holds three very different textures: folded quartzite ridges with long linear features, the dense Gurugram street grid, and smooth crop parcels. The scene was selected by our code, not by eye. Scene-wide cloud cover was near zero for almost all candidates, but inside our area the cloudy or hazy share went up to 97.2 percent because of winter smog. The chosen scene has only 0.08 percent flagged pixels.

What we did



Spatial domain: slide a kernel

$g(x, y) = \sum_{s,t} w(s, t)\, f(x - s,\, y - t)$

mean, Gaussian, median; HPF, Laplacian, Sobel, Prewitt; high-boost

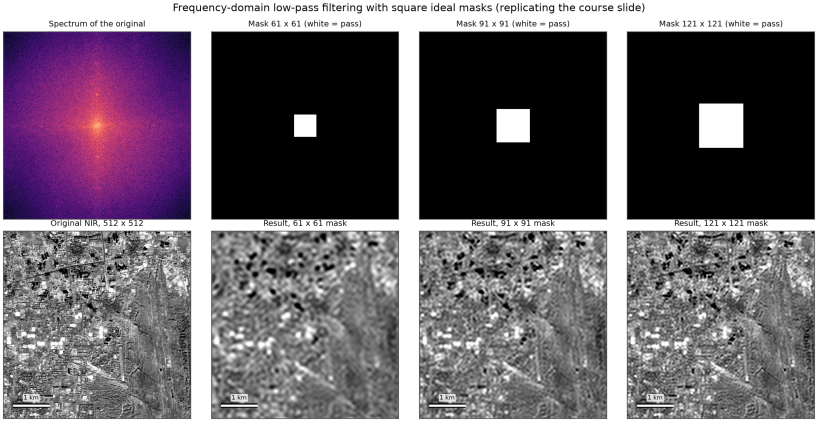
Frequency domain: mask the spectrum

$g = \mathcal{F}^{-1}\{H(u, v) \cdot \mathcal{F}\{f\}\}$

ideal, Butterworth, Gaussian masks; HPF mask $= 1 - H$

The pipeline has four steps. We implemented every filter in plain NumPy and checked each against SciPy; the largest difference is below ten to the minus five. Filters come in two routes. In the spatial domain a small grid of weights slides over the image. In the frequency domain we take the Fourier transform, multiply by a mask that keeps some frequencies and removes others, and transform back. Then five experiments measure noise removal, classification, edges, spectral change and speed, and the decision matrix picks the winner for each task.

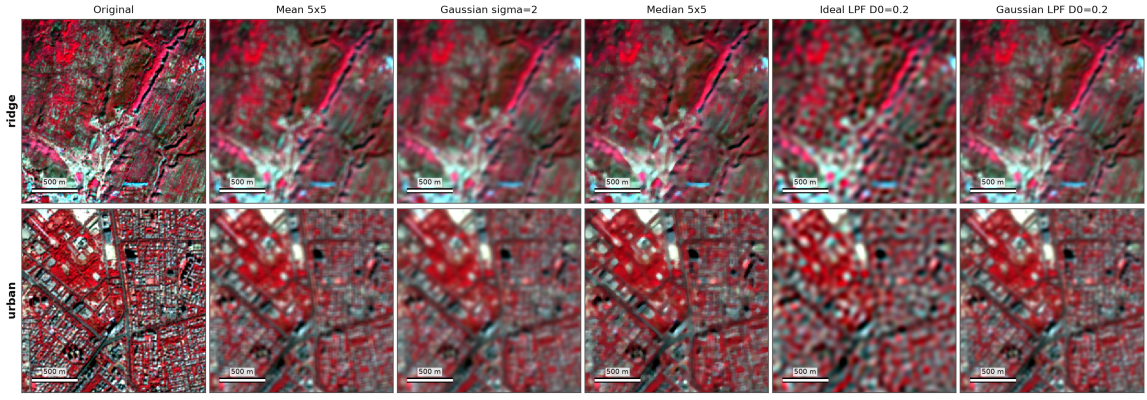
Frequency domain: the lecture example, reproduced on our scene



Smaller mask ⇒ fewer frequencies kept ⇒ blurrier image, with ripples

This is the example from the course slides, with square ideal masks of 61, 91 and 121 pixels, applied to a 512 by 512 crop of our scene. The top row shows the spectrum and the masks: white means pass. The bottom row shows the result. The smallest mask keeps the fewest frequencies and gives the blurriest image. Note the ripple texture in all three; we come back to that in two slides.

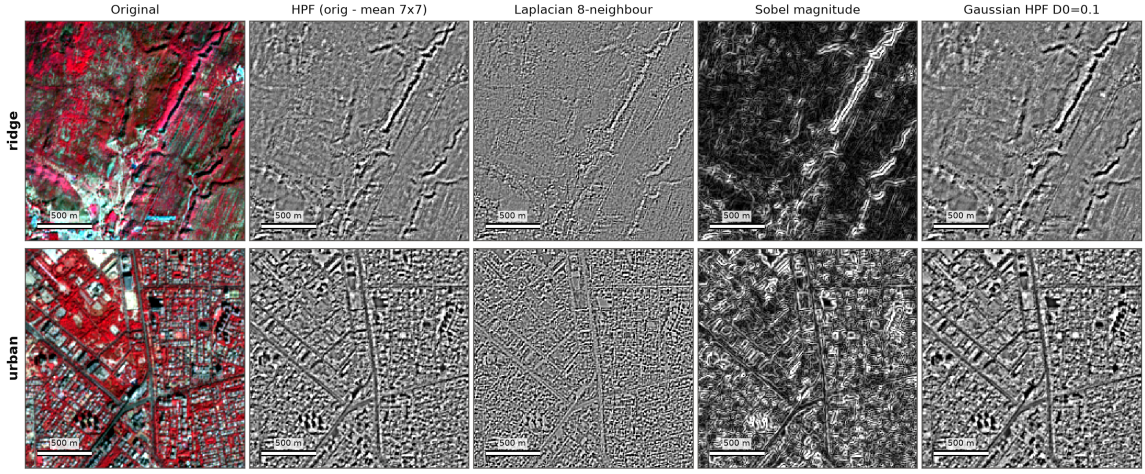
Low-pass filters smooth texture and blur boundaries



Broad ridge shape survives; street grid and field edges are lost. Median keeps edges best.

Two 2 kilometre windows, ridge on top and Gurugram below, all with the same colour stretch. The mean and Gaussian blur the street grid. The median keeps straight edges a little sharper because it picks an existing value instead of averaging across the edge. In the frequency domain, the ideal mask leaves a ripple texture while the Gaussian mask looks just like the spatial Gaussian, which is expected because they are the same filter.

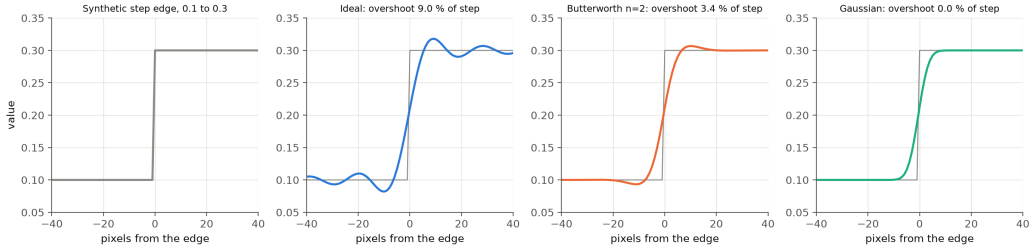
High-pass filters keep only the edges



Flat ground turns grey. Sobel and the 7×7 HPF trace ridge crests; the Laplacian picks up speckle.

These are pure high-pass layers of the near-infrared band. Uniform areas go to mid-grey and only the changes remain. The 7 by 7 HPF and the low-cutoff frequency HPF bring out the ridge crests, and Sobel shows edge strength in any direction. The Laplacian reacts most to the finest speckle. In practice an analyst adds these layers back to the image, which is the high-boost filter we test later.

A sharp cut in frequency causes ringing



Mask ($D_0 = 0.1$)	Ideal	Butterworth	Gaussian
Overshoot at a step edge	**9.0%**	3.4%	0.0%

Never use the ideal mask. Gaussian or Butterworth masks are safe.

Real scene texture hides the ripples, so we pushed a clean step edge through the same filter code. The ideal mask cuts frequencies off sharply, and that always produces ripples beside edges, the Gibbs effect. It overshoots by 9.0 percent of the step, close to the nine percent theory predicts. Butterworth overshoots 3.4 percent and Gaussian 0.0.

E1 Noise: the right filter depends on the type of noise

Random (Gaussian) noise

Noisy input	33.98 dB
Gaussian $\sigma = 1$	**35.56 dB**
Mean 3×3	35.52 dB

13 of 18 filters made it *worse*

Salt-and-pepper noise

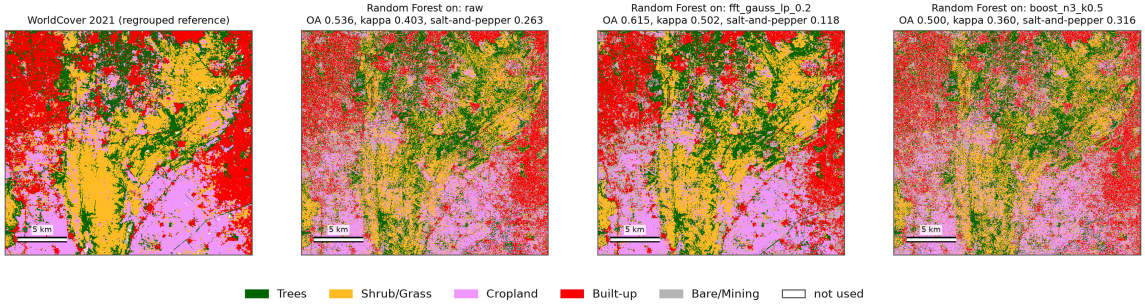
Noisy input	21.01 dB
Median 3×3	**36.49 dB**
Gaussian LPF $D_0 = 0.2$	31.06 dB

median beats the best linear filter by 5.42 dB

PSNR against the clean image, higher is better. Light filters only: large kernels remove detail along with noise.

We added two kinds of noise to a clean crop and measured how close each filtered result came to the clean image. With random noise only the lightest smoothing helped, and most filters made things worse because they removed real detail too. With salt-and-pepper noise, the median was in a class of its own: a mean or Gaussian spreads each spike over its neighbours, while the median simply throws it away.

E2 Classification: smoothing gives more accurate, cleaner maps



Cohen's κ on held-out blocks: raw 0.403 $\rightarrow$ **best LPF 0.502**; best high-boost 0.360

Smoothing removes within-class variation (roof types, shadows) before the classifier sees it.

We trained a Random Forest on the four bands plus NDVI, with WorldCover as the reference, after each filter. Training and test pixels come from separate 128 by 128 pixel blocks, because neighbouring pixels are near-copies and a random split would inflate accuracy. Every low-pass input beat the raw bands. The best raised kappa from 0.403 to 0.502 and cut the map speckle index from 0.263 to 0.118. High-boost made it worse, at best 0.360.

E3, E4: high-pass finds edges, but the output is no longer reflectance

Edges and lineaments (NIR)

	Original	Best
Boundary F1	0.675	**0.742** high-boost
Ridge coherence	0.510	**0.536** Prewitt

Smoothing lowers boundary F1 (0.247 for mean 5×5)

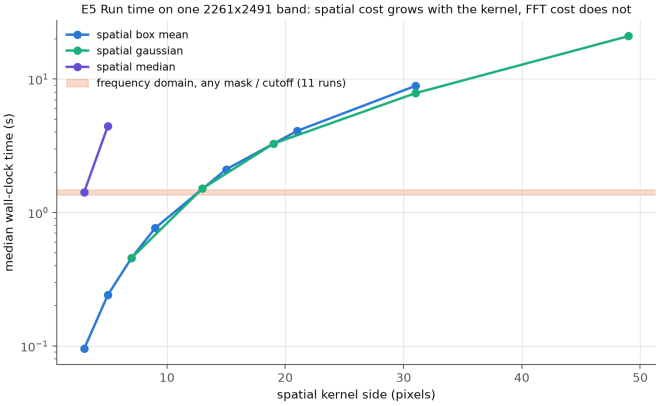
Cost: change to the spectrum

	Spectral angle
High-boost $n = 3$, $k = 0.5$	1.20°
Median 3×3	2.22°
Gaussian $\sigma = 1$	2.30°
Pure HPF layers	$\approx 89°$

Use HPF products as *extra layers* beside the bands, never in place of them.

For edges the picture flips. Canny edges from high-boost images match the WorldCover boundaries better than the original, while smoothing removes edges. For the ridges we measured orientation coherence, which is high when local gradients line up; the gradient operators Prewitt and Sobel gave the clearest linear structure. The cost is on the right: light filters change the spectrum by one or two degrees, but a pure high-pass layer is no longer a reflectance image, so it must stay a separate layer.

E5 Speed: FFT cost is flat, convolution cost grows with the kernel



One full band, 2261×2491 pixels:

- Spatial 3×3: 0.10 s
- Spatial 49×49: 20.96 s
- Any FFT mask: 1.36–1.48 s

Break-even near a 13-pixel kernel

Small kernels: convolve. Large kernels or many cutoffs: use the FFT.

Spatial cost grows with the square of the kernel side; the Fourier route costs about the same whatever the mask. Below a kernel of about 13 pixels convolution is faster, above it the FFT wins. Since the filters that worked best are small, convolution is usually the practical choice.

The answer: one best filter per application

Application	Best filter	Evidence (best vs. no filter)
Noise removal, random noise	Gaussian $\sigma = 1$	PSNR (dB): **35.56** vs 33.98
Noise removal, impulse noise	Median 3×3	PSNR (dB): **36.49** vs 21.01
Land-cover classification	Gaussian LPF $D_0 = 0.2$	Cohen's κ: **0.502** vs 0.403
Lineament mapping (ridges)	Prewitt	ridge coherence: **0.536** vs 0.510
Urban boundary delineation	High-boost $n = 3$, $k = 0.5$	built-up edge F1: **0.545** vs 0.543
Visual interpretation	High-boost $n = 3$, $k = 1.0$	NIR mean gradient: **0.0417** vs 0.0245

Low-pass when pixel values feed a calculation; **High-pass** when edges and structure are the point.

This table is generated by our code from the experiment results. Each row uses one measure. Some gains are large, like the median on impulse noise or the Gaussian for classification. Some are small, like urban boundaries, and we report them as small. The pattern is clear: low-pass filtering wins wherever per-pixel values are used for a calculation, and high-pass filtering wins wherever the task is about where things change.

Take-home message

Quantitative work

classification, indices, change detection

⇒ light **Gaussian** ($\sigma \approx 1.5$–2 px)
⇒ **median** 3×3 for impulse noise

Interpretation and mapping

visual analysis, lineaments, boundaries

⇒ **high-boost** $k = 0.5$–1 for display
⇒ **Sobel / Prewitt** for lineaments

Avoid

the ideal frequency mask (ringing), and heavy smoothing (detail goes before noise)

Live demo: `python -m rsfilter demo` (offline, 512×512 sample, about one minute)

Thank you. Questions?

So, which technique is most suitable? For anything quantitative, use a light low-pass filter: a small Gaussian, or a 3 by 3 median when there is impulse noise. For interpretation and structural mapping, keep the bands and add a high-pass product beside them: high-boost for display, Sobel or Prewitt for lineaments. Never use the ideal mask. We will now run the demo, which repeats the pipeline on the sample crop in about a minute.

References

Josef Bigün and Gösta H. Granlund, *Optimal orientation detection of linear symmetry*, Proceedings of the First International Conference on Computer Vision (London), 1987, pp. 433–438.

Leo Breiman, *Random forests*, Machine Learning **45** (2001), no. 1, 5–32.

John Canny, *A computational approach to edge detection*, IEEE Transactions on Pattern Analysis and Machine Intelligence **8** (1986), no. 6, 679–698.

M. Drusch, U. Del Bello, S. Carlier, O. Colin, V. Fernandez, F. Gascon, B. Hoersch, C. Isola, P. Laberinti, P. Martimort, A. Meygret, F. Spoto, O. Sy, F. Marchese, and P. Bargellini, *Sentinel-2: ESA's optical high-resolution mission for GMES operational services*, Remote Sensing of Environment **120** (2012), 25–36.

European Space Agency, *Sentinel-2 Level-2A Product Specification and Sen2Cor Algorithm Theoretical Basis Document*, ESA Sentinel Online.

Rafael C. Gonzalez and Richard E. Woods, *Digital image processing*, 4th ed., Pearson, New York, 2018.

John R. Jensen, *Introductory digital image processing: A remote sensing perspective*, 4th ed., Pearson, Glenview, IL, 2015.

Thomas M. Lillesand, Ralph W. Kiefer, and Jonathan W. Chipman, *Remote sensing and image interpretation*, 7th ed., Wiley, Hoboken, NJ, 2015.

Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli, *Image quality assessment: from error visibility to structural similarity*, IEEE Transactions on Image Processing **13** (2004), no. 4, 600–612.

Daniele Zanaga, Ruben Van De Kerchove, Dirk Daems, Wanda De Keersmaecker, Carsten Brockmann, Grit Kirches, Jan Wevers, Oliver Cartus, Maurizio Santoro, Steffen Fritz, et al., *ESA WorldCover 10 m 2021 v200*, Zenodo, 2022.